

Case Manager

An AI-Assisted Mobile Application for
Health Record Organization & Management

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Advisory Committee: Dr. Kathy Gray | Dr. Edward Andrew Roualdes (Chair)

What is Case Manager?

- A voice-first, AI-powered mobile health management application
- Designed for senior citizens and individuals with special needs
- Consolidates medications, appointments & medical records in one place
- Hands-free natural language voice interaction — no typing required
- AI responses grounded in the user's own personal health data
- Cross-platform: iOS, Android, and Web

Why We Built It

- 703M+ people aged 65+ globally — projected to double by 2050 (UN, 2023)
- Seniors struggle to manage complex health regimens consistently
- Missed medications and appointments lead to worse health outcomes
- Caregivers face significant administrative burden tracking health info
- No unified, voice-first solution combining AI with personal health data existed
- Goal: reduce cognitive load and improve health independence

The Problem

- Managing multiple medications dosages, timings, and instructions is cognitively demanding for senior citizens
- Seniors frequently miss doses or appointments due to memory and coordination challenges
- Increased stress for both individuals and caregivers leads to poorer health engagement and outcomes
- Health data is fragmented across paper records, multiple apps, and verbal memory no single source of truth
- Traditional reminder apps still require significant manual setup and do not adapt to individual user context
- No unified, voice-first solution existed that combines AI with a user's personal health data

Case Manager Solves This By...

Voice-Activated AI

Ask health questions naturally — no typing, no manual input required

Personal Health Data

AI understands and responds using the user's own Firestore medical records

10 Record Categories

Unified hub for medications, conditions, labs, allergies, providers and more

Safety Validation

Multi-agent safety layer validates every AI response before delivery

Emergency Detection

Automatic symptom detection and medical disclaimers on every query

PDF Export

Shareable clinical document for provider communication via native share sheet

Evolution of AI in Healthcare

- 1975: MYCIN at Stanford — AI diagnosed bacterial infections at specialist level
- 1990s: Rise of EHRs laid the data foundation for clinical AI
- 2000s: NLP chatbots began handling medication reminders & patient engagement
- 2015+: Machine learning dramatically accelerated clinical AI adoption
- 2024: Generative AI now embedded in EHR workflows across leading health systems

What Research Tells Us

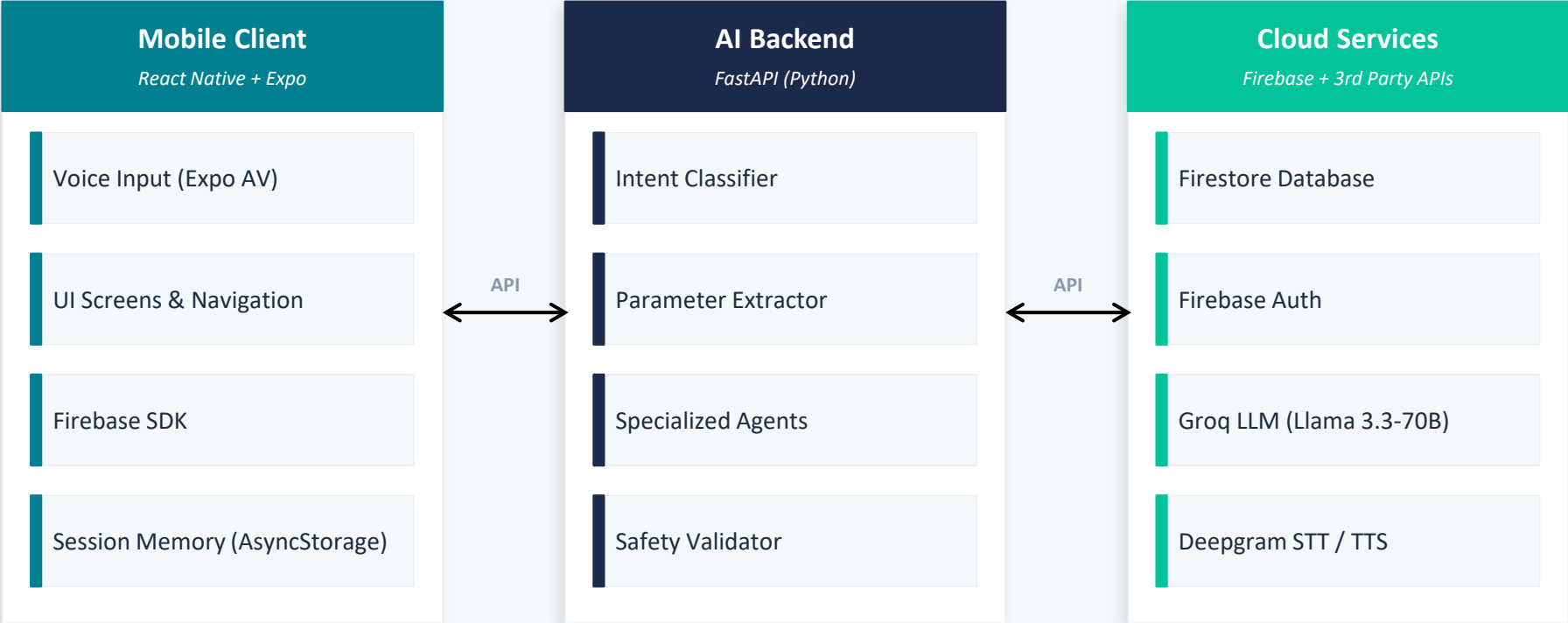
- 81% of adults aged 50–80 have used at least one digital health technology (Anderson & Perrin, 2017)
- 89.3% of seniors value appointment reminders most in AI tools (Wolfe et al., 2025)
- AI reduces hospital readmissions by ~15% (Medwave, 2024)
- Wearable AI monitoring cuts emergency admissions by 20% (Li et al., 2025)
- NLP documentation automation cuts admin time by 40% (Datagrid, 2025)

Limitations of Existing Tools

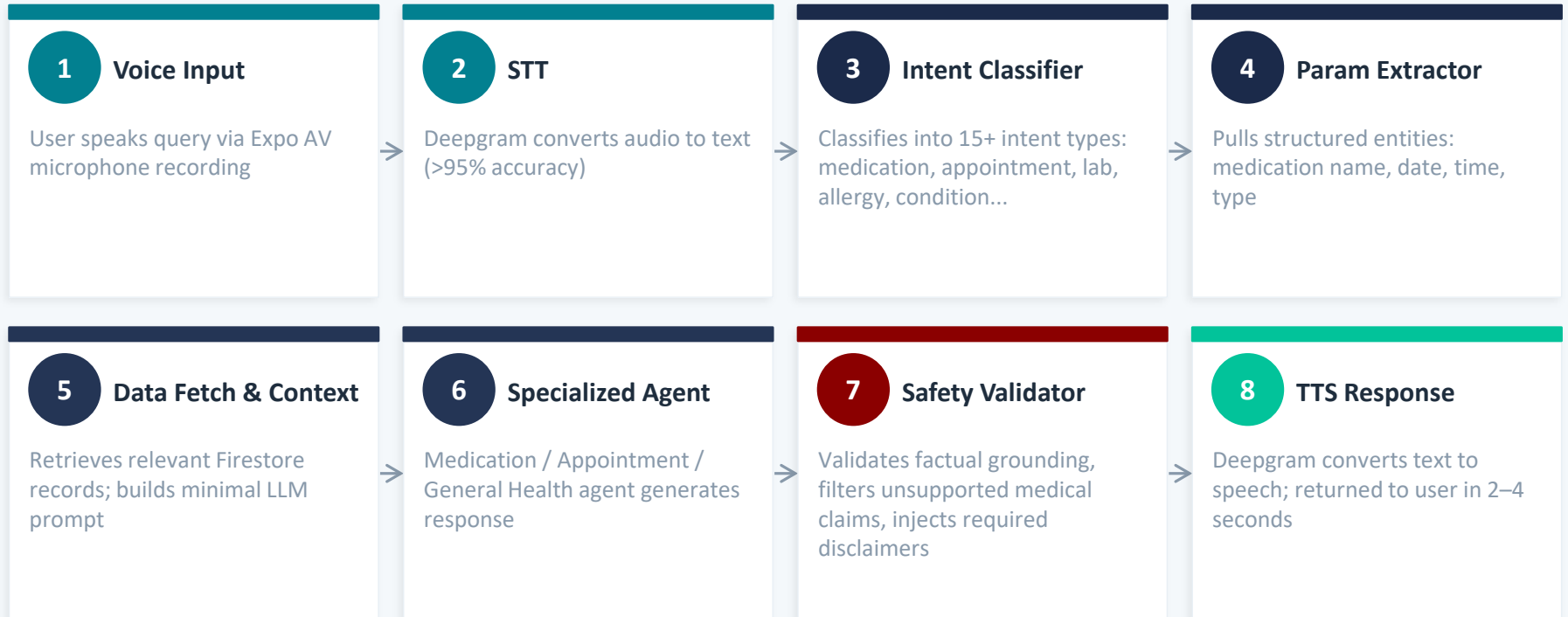
- Most health apps require significant manual data entry from users
- To our knowledge, no unified voice-first personal health assistant existed at time of development
- AI chatbots are rarely grounded in a user's own personal health records
- Safety-validated LLM assistants in the health domain remain scarce
- Senior citizens remain largely underserved by current mobile health apps
- No single app previously combined records + AI + voice + mandatory safety validation

What Case Manager Adds

- Voice-first interface — Reduced manual typing
- AI responses grounded in user's own Firestore records
- Mandatory safety validation layer on every AI response
- 10 medical record categories in one unified app
- Emergency symptom detection + allergy cross-referencing
- Exportable PDF for sharing with healthcare providers



Voice Input → STT (Deepgram) → Intent Classification → Agent Pipeline → Safety Validation → TTS Response



Technologies Used

Frontend

React Native + Expo, Expo Router, Expo AV, AsyncStorage

Backend

FastAPI + Uvicorn, Groq Llama 3.3-70b, Deepgram SDK

Database

Firestore (NoSQL) with real-time listeners, Firebase Auth

Data & Testing

Sample health data seeded per user; 3-level testing: frontend, backend endpoint, end-to-end integration

Key Design Decisions

- Server-side AI processing — keeps API keys secure and centralizes safety policy
- Modular agent architecture — each agent refined independently without restructuring
- Per-user Firestore namespace — prevents cross-user data access by design
- Session memory via AsyncStorage — persists across tabs, resets on app close for privacy
- Iterative dev approach — core features first, AI integration second, safety last layer validated
- Trade-off accepted: cloud-dependency over offline capability for v1 scope



Voice AI Assistant

Successfully implemented natural language query handling grounded in user health records. Hands-free, context-aware.



Medication Management

Implemented dosage tracking, timing reminders, mark-as-taken, and automatic allergy cross-referencing.



Appointment Scheduling

Successfully delivered calendar event management via voice and manual entry across all platforms.



Medical Records Hub

10 structured categories implemented with full CRUD: allergies, conditions, labs, prescriptions, immunizations & more.



Safety Mechanisms Validated

Emergency symptom detection, factual grounding validation, and auto medical disclaimers confirmed in testing.

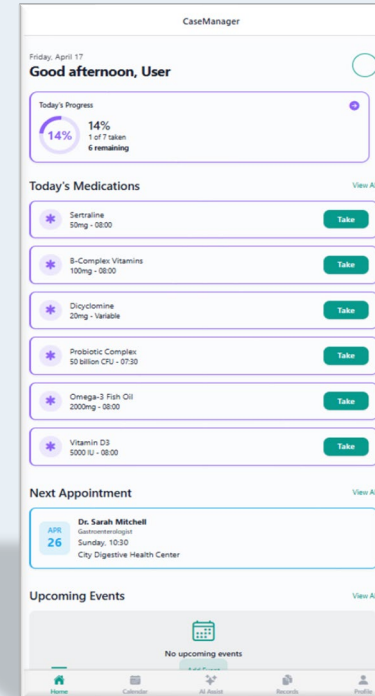


PDF Export

Successfully generates and shares a comprehensive clinical document via native share sheet across iOS, Android, and Web.

User Interface — Login & Dashboard

- Secure Firebase Authentication — email/password login with email verification workflow
- Password reset support with Firebase Auth email flow
- Main dashboard displays daily medication schedule at a glance
- Upcoming appointments shown with date, time and type
- Health status summary for centralized daily monitoring
- Single-tap tab navigation to AI assistant, records hub, and dashboard

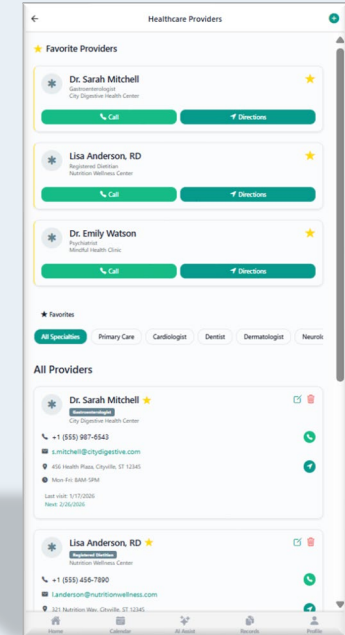
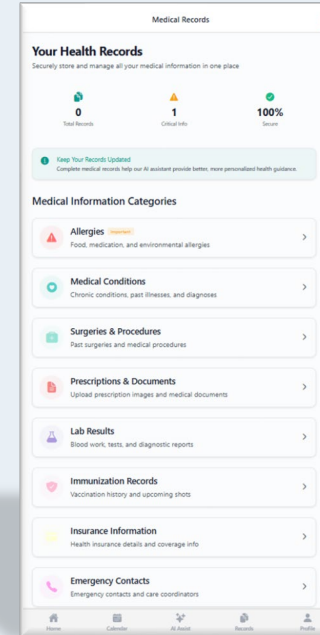


[Login / Dashboard Screen]

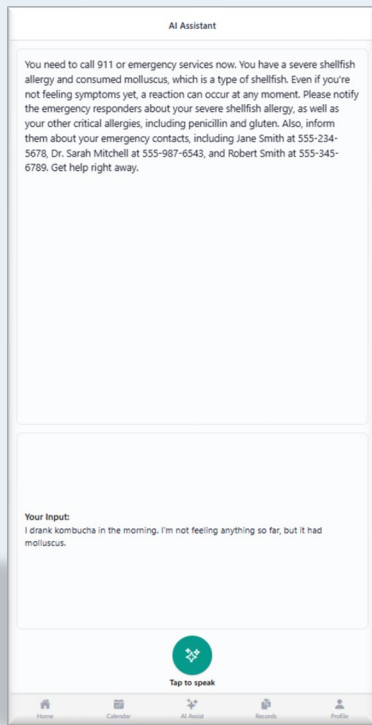
User Interface — Medical Records Module

05 FEATURES & DEMO

- 10 structured record categories: allergies, conditions, medications, immunizations, lab results, prescriptions, surgeries, providers, insurance, emergency contacts
- Full CRUD operations — add, view, edit, and delete any record
- Active vs. managed conditions clearly distinguished for continuous monitoring
- Healthcare providers stored with contact details, specialty, and location
- Real-time synchronization across devices in under 500ms



[Medical Records / Conditions Screen]



[AI Assistant Screen]

Key Outcomes Achieved

- Successfully classified 15+ distinct intent types with confidence scoring across all test queries
- Maintained conversational context across an average of 6 message exchanges per session
- Allergy cross-referencing and emergency symptom detection validated across all test cases
- System operational on iOS (v14+), Android (API 21+), and modern web browsers

>95%

STT Accuracy

2–4s

Response Time

15+

Intent Types

<500ms

Sync Latency

10

Record Categories

3

Platforms

Technical Constraints

Internet Dependency

All AI features and database operations are cloud-dependent. No offline fallback exists — the app is non-functional without an active internet connection, which is a significant constraint for a health management tool.

English-Only

Deepgram STT and all LLM prompt templates are configured exclusively for English. This substantially limits the app's reach among non-English-speaking seniors and caregivers.

Backend Dependency

The FastAPI server must be actively deployed and reachable for the AI assistant to function. Any server downtime directly disables the core differentiating feature of the application.

Latency for Critical Use

Cloud-based AI processing introduces 2–4 seconds of round-trip latency, making the system unsuitable for truly real-time or emergency-critical health alerts.

Scope & Compliance Constraints

Sample Data Only

The system uses seeded sample data and is not HIPAA-compliant. Real patient data cannot be handled without a fully compliant data pipeline, security audit, and regulatory review.

No EHR Integration

The app cannot read from or write to hospital electronic health record systems. Achieving this would require HL7 FHIR compliance, institutional partnerships, and significant regulatory oversight.

Diagnostic Scope

Diagnostic and prescriptive capability is intentionally out of scope due to safety and regulatory considerations. The system is constrained to informational guidance and record summarization only.

No Real User Testing

The system was not evaluated with actual members of the target demographic — senior citizens or individuals with special needs. This is a key limitation on the validity and generalizability of results.

01 HIPAA Compliance & Real Data

Establish a secure, fully compliant health data pipeline before real-world deployment with actual patient data

02 Offline-First Capabilities

Implement local database sync so medication schedules and emergency contacts work without internet

03 Multi-Language Support

Plan internationalization from the architecture phase — Deepgram and LLM prompts need language-specific configs

04 EHR Integration

Connect with electronic health records via HL7 FHIR standards for seamless provider-patient data exchange

05 Expanded Accessibility

Add advanced visual aids, customizable inputs, and onboarding for users with cognitive or sensory impairments

06 Clinical Validation & Trials

Real-world user testing with target demographic was not conducted — a key limitation. Future work should involve healthcare professionals and structured trials.

- [1] United Nations Department of Economic and Social Affairs. (2023). World population ageing 2023. United Nations.
- [2] Anderson, M., & Perrin, A. (2017). Tech adoption climbs among older adults. Pew Research Center.
- [3] Wolfe, C., Bhatt, A., & Hyman, J. (2025). Older adults' satisfaction with AI caregiving chatbot features. Journal of Gerontological Nursing.
- [4] Medwave. (2024). Predictive analytics and hospital readmission reduction in older adults. Medwave.
- [5] Li, W., Luo, J., & Chen, X. (2025). Wearable AI monitoring and emergency admission reduction. Journal of Medical Internet Research.
- [6] Datagrid. (2025). NLP-driven documentation automation in healthcare case management. Datagrid Technical Report.

Thank You

Questions & Discussion

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